

To Generate or Not to Generate: Using an LLM to Train a Drug-Drug Interaction Extractor

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Abstract

Annotation is the usual bottleneck in building a relation extraction system, and a large language model (LLM) that could write labelled training data would remove it. We tested this for drug-drug interaction extraction on DDI-2013 by fine-tuning BiomedBERT on sentences an LLM wrote from structured specifications that fix each pair’s label in advance. The best of five generators reached 0.528 micro-F1 on the test set, against 0.806 for the same classifier trained on the human-annotated training set. Much of the gap came from shortcuts in the generated data. In the first generator the number of drugs in a sentence predicted its label, and in the fourth the sentence’s construction determined it, so a classifier could learn the label without reading the relation. On the same 2,402 real sentences, replacing human labels with LLM labels cost 0.126 F1, and replacing the real sentences with generated ones cost a further 0.152, although the generated set had 2.3 times as many training pairs. The LLM’s labels over-predicted interactions, and 36% of its positive labels are NONE in the gold annotation. A second LLM pass that checks each positive label and demotes those it rejects removed mostly errors, since 91% of the labels it rejected are NONE in gold, against 20% of those it accepted. Trained on these checked labels with no human annotation, the classifier reached 0.728, 90% of full human supervision and within 0.02 of 1,000 human-annotated sentences. With 1,000 human-annotated sentences, 42% of the training set, and checked labels on the rest, it reached 0.773, 96%. Adding generated text to LLM-labelled real text lowered F1 at every annotation budget we tested. For this task an LLM is better spent labelling and checking real text than writing new text.

A companion notebook, `tutorial/ddi_tutorial.ipynb` in github.com/notgwilym/ddi-synth, runs the analyses of this report on samples of the project’s own data. Sections 2 to 7 follow the notebook’s sections 1 to 6 in the same order. Boxed examples point to the cells that show the sentences and pairs behind a result, and Appendix A maps the two documents.

1 Introduction

Relation extraction systems learn from annotated examples, and annotation is usually the slowest and most expensive part of building one. Drug-drug interaction extraction is a demanding case. The DDI-2013 corpus (Herrero-Zazo et al., 2013) labels every pair of drug mentions in a sentence with one of five relations, under detailed guidelines that as well as defining the labels also decide which mention takes part in an interaction, and how denials and contradictory statements are treated. Annotating a new corpus to that standard requires trained annotators.

Large language models (LLMs) offer a way around this, but not by doing the extraction themselves. On biomedical information extraction, BERT-sized models fine-tuned on annotated data still outperform GPT-3 with in-context examples (Jiménez Gutiérrez et al., 2022), and a comparison of four recent LLMs on twelve biomedical benchmarks found traditional fine-tuning ahead of zero- and few-shot prompting on most tasks (Chen et al., 2025). A small fine-tuned

model is also far cheaper to run over a large literature. The question is therefore whether an LLM can supply the training data that a small model needs.

There are two ways it can. The first is to generate labelled text. Tang et al. (2023) trained models for biomedical entity and relation extraction on sentences written by ChatGPT, Xu et al. (2024) added clinical knowledge to generation prompts across eighteen clinical datasets, and Josifoski et al. (2023) generated text for given relation triples, the reverse of the extraction task, and outperformed the previous state of the art in closed information extraction by a wide margin. Delmas et al. (2024) found that relation extraction models for natural-products literature improved when trained on generated abstracts rather than the original noisily labelled data. The second way is to keep real text and let the LLM label it. Wang et al. (2021) found that labels from GPT-3 matched the downstream performance of human labels at 50% to 96% lower cost, and Gilardi et al. (2023) found ChatGPT more accurate than crowd workers on several annotation tasks. Tan et al. (2024) survey both approaches.

Few studies compare the two on the same task. Ding et al. (2023) did so for several tasks with GPT-3 and found that labelling real data suited tasks with few labels, while generating data suited tasks with many. Kazemi et al. (2025) compared LLM-generated labels and LLM-generated text against gold annotation for cyberbullying detection.

This report makes that comparison for drug-drug interaction extraction on DDI-2013, fine-tuning BiomedBERT (Gu et al., 2021) on every kind of training data. It makes four contributions. The first is a controlled comparison on the same 2,402 real sentences, which separates the cost of model labels from the cost of generated text. The second is an account of why generated text scored lower, through shortcuts that let a classifier read a sentence’s label from its form, together with two diagnostics that detect them before any text is generated. The third is a check in which the model reviews its own positive labels, measured against the gold annotation. The fourth is an evaluation of how to spend a limited annotation budget, chosen on a validation set and tested once. Across five generator designs, the best generated text reached 0.528 F_1 on the test set, against 0.806 for human labels. Checked model labels reached 0.728 with no human annotation and 0.773 when combined with 1,000 human-labelled sentences. A companion notebook reproduces the analyses on samples of the project’s data.

2 Task, Data and Evaluation

Drug-drug interaction extraction on DDI-2013 (Herrero-Zazo et al., 2013) takes a sentence whose drug mentions are already marked and asks, for every unordered pair of mentions, whether the sentence states an interaction between them and of which kind. There are five labels. MECHANISM marks a pharmacokinetic change, EFFECT a clinical or pharmacodynamic consequence, ADVISE a recommendation about taking the two together, INT an interaction stated without detail, and NONE everything else. A sentence with k marked mentions yields $\binom{k}{2}$ pairs, and each pair is labelled separately. Labels attach to mentions rather than to drug names, so a drug named twice forms two mentions, and the annotation guidelines usually let only one of them take part in a given interaction. About five pairs in six are NONE, so every score in this report is micro-averaged F_1 over the four positive labels, which does not reward predicting NONE.

Example 2.1

Tutorial §1 prints a real sentence in which Clozapine and carbamazepine are each named twice. Its six pairs show labels attaching to mentions. ADVISE links the first two mentions, MECHANISM links the last two, and every pair across the two clauses is NONE.

→ TUTORIAL §1

We split the corpus’s training directory at document level into a training set, a development set and a validation set in the proportions 70:15:15, and keep the official test directory as the

test set (Table 1). The development set guided the design of the generators, the validation set chose between configurations, and the test set was used once, for the configurations chosen on validation. The document identifiers of each split are recorded, so the split can be rebuilt on any machine.

Table 1: The four data sets.

Set	Pairs	Used for
Training	14,758	training every classifier
Development	4,243	designing and comparing generators
Validation	6,276	choosing between configurations
Test	6,744 (976 positive)	one evaluation of the chosen configurations

We do not apply the negative instance filtering common in published work on this corpus, which removes trivially negative pairs, such as two mentions of the same drug, before training and evaluation. Published scores on the benchmark range from about 65 to 90, much of that spread reflecting preprocessing, and our scores are not comparable with them. DDI-2013 consists of drug labels from DrugBank and abstracts from MedLine, so every result here concerns literature mining.

Every configuration fine-tunes the same classifier for internal consistency, BiomedBERT (Gu et al., 2021), with learning rate 2×10^{-5} , batch size 32 and maximum sequence length 256. Each configuration trains for three epochs, or for 500 optimisation steps if three epochs would give fewer. Training sets in this study range from a few hundred pairs to tens of thousands, and without this minimum the smallest would receive about 60 steps, after which the classifier predicts NONE for every pair. We report the mean of three to five seeds, and give the difference between two configurations with its number of standard errors,

$$SE(\bar{a} - \bar{b}) = \sqrt{\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}}, \quad (1)$$

where s and n are the standard deviation and number of seeds of each configuration. Comparisons are only made between scores on the same set.

3 Writing Labelled Sentences

3.1 Specifications

A generated sentence begins as a *specification*, which lists the drug names the sentence must contain and the relation, if any, that each pair among them must express. The names are drawn at random from 14,932 DrugBank names and 425 WHO-ATC drug groups, so most combinations make no clinical sense. The task is to recognise what a sentence states, not whether the interaction is real, and random pairs stop a classifier from learning which drug names tend to interact. In early versions, replacing every drug name with a placeholder lowered the F_1 of human-labelled data by 0.021 and raised that of generated data by 0.011 on the development set, so drug identity contributed little to either. Whether pairs known to interact would help was not tested. The specification is rendered as a prompt and sent to **gpt-oss-120b** at temperature 0.9. A second, deterministic step finds each drug in the sentence the model wrote, tags every pair, and copies each pair’s label from the specification. Sentences that omit a required drug are discarded. Each label is therefore the one requested, not one read from the sentence, and if the model writes a sentence that does not express the requested interaction, the label is wrong.

Example 3.1

Tutorial §2 prints a G5 specification, the prompt rendered from it, the sentence the model wrote and the pairs with their labels. Pairs the specification does not name are NONE, so one generated sentence gives negative pairs as well as positive ones.

→ TUTORIAL §2

Five generators were trained with several seeds and scored on the final evaluation sets. We call them G1 to G5 (Table 2). Earlier prompt designs, single sentences with stated negatives and multi-sentence clinical vignettes, were revised a dozen times, and only two of those revisions were ever trained, scoring 0.316 and 0.247 on the development set.

Table 2: The five generators. The repository identifies them by their original version numbers.

Generator	Repository	Design
G1	v13	a sentence expressing a given relation between two given drugs
G2	v14	adds drugs outside the relation, so that sentences contain NONE pairs
G3	v15	draws sentence content to match the corpus
G4	v17	builds each sentence from one of nineteen frames taken from the annotation guidelines
G5	v18	separates label-carrying assertions from label-neutral modifiers

3.2 The pair-count shortcut

All scores in this section are on the development set (Table 3). G1 wrote what it was asked for, a sentence in which two given drugs interact, so almost every sentence had exactly one pair and nine in ten of those pairs were positive. Its label could be predicted from the number of drug mentions. Sentences containing one pair were 74.8% positive, and sentences containing two or three pairs were 0.2% positive. In the corpus, a sentence of any size can contain both positive and NONE pairs. G2 added drugs outside the relation, which removed this dependence. Its precision rose from 0.184 to 0.288, and its recall fell from 0.588 to 0.476.

Example 3.2

Tutorial §2.2 gives a rule that sees only the number of drugs in a sentence. Firing on sentences with two drugs gets 0.91 of G1’s sentence labels right and 0.43 of the real ones, and the table that follows shows the positive rate by number of drugs for every generator.

→ TUTORIAL §2.2

3.3 The construction shortcut

G4 built each sentence from one of nineteen named constructions, called frames, taken from the annotation guidelines. Each frame always produced the same sentence label. A classifier trained on G4 could therefore learn which kinds of sentence are positive instead of learning to read whether two drugs interact. We measure how much a construction gives away with the *uncertainty coefficient* (Theil, 1970). With Y the sentence label and F the construction,

$$U(Y | F) = \frac{H(Y) - H(Y | F)}{H(Y)}, \quad H(Y) = - \sum_y p(y) \log_2 p(y), \quad (2)$$

which is 1 when the construction decides the label and 0 when it says nothing about it. Suppose half of a set of sentences use a mechanism construction and half a denial, and half of all sentences are positive. Then $U = 1$ if every mechanism sentence is positive and every denial negative, $U = 0.28$ if each construction is four-fifths one label, and $U = 0$ if each is half positive. The target is not zero, because real constructions do carry information about labels.

A sentence stating that one drug raises the levels of another is usually positive, and a denial is usually NONE. The target is the corpus’s own value, measured on 300 corpus sentences that `gpt-oss-120b` classified by construction at temperature 0. Because the coefficient needs only specifications, a generator can be checked with it before any text is generated. For G4 it is 1.000.

A construction that decides the label only matters if the classifier can see it, and the classifier sees the sentence, not the specification. To test this we masked every drug name in the G4 sentences and trained a bag-of-words logistic regression to recover the frame, scored by three-fold cross-validation (Tutorial §2.3). It recovered the frame for 0.961 of sentences, where guessing would get one in nineteen. The wording therefore gives away the frame, and the frame gives away the label. A representative G4 sentence, with its drug mentions marked, shows two kinds of cue.

`Pelabresib`, on which `Ravoxertinib` acts and raises hepatic clearance about threefold as measured by plasma concentrations, is administered, while `LetibotulinumtoxinA` is used if the first choice is unsuitable.

The phrase *measured by* is among the strongest features the bag-of-words model associates with the mechanism frame. The phrase *used if the first choice is unsuitable* is one of seven fixed descriptions G4 gave to every drug outside the relation. These descriptions were introduced to stop the model simply listing the extra drugs, and they became a cue of their own, since each one marks its drug’s pairs as NONE without any pharmacology. They are the most repeated 4-grams in G4’s sentences, where the most repeated 4-grams in real text are lists of drugs.

Example 3.3

Tutorial §2.3 builds this analysis from one G4 sentence. It tabulates every frame against sentence label, computes the coefficient by hand and on toy generators, runs the bag-of-words test, and compares each construction’s positive rate in G4 with its rate in the corpus.

→ TUTORIAL §2.3

The corpus comparison also shows which constructions carry labels. Appositive and coordinate constructions appear in real sentences but are never their main clause, because they package an assertion rather than make one. G4 treated them as frames of their own and made appositive sentences positive every time, where the corpus rate is about four in ten.

3.4 Separating assertions from modifiers

G5 splits the specification into two parts. *Assertions* bind a pair of drugs and carry a label, and a sentence draws one or two of seventeen, with a second assertion present with probability 0.55 and negative with probability 0.55. *Modifiers*, such as headings, repeated mentions and dose details, carry no label and are drawn independently from eight. A positive assertion can now appear next to a negative one, and any assertion can appear with any modifier, so no single part of a specification decides the sentence’s label. Appendix B lists every assertion, modifier and variant with a sentence G5 wrote from it. G5’s uncertainty coefficient is 0.458, against 0.464 for the corpus. On the development set its F_1 rose from G4’s 0.391 to 0.486 (Table 3, Figure 1). The gain was in recall, which rose from 0.403 to 0.628, while precision rose only from 0.381 to 0.398.

Table 3: Development scores of each generator (standard deviation across seeds in parentheses). G1 is scored on the development set less three pairs whose overlapping spans could not be marked.

Generator	F_1	P	R	Seeds
G1	0.279 (0.011)	0.184	0.588	3
G2	0.359 (0.027)	0.288	0.476	3
G3	0.333 (0.010)	0.272	0.430	3
G4	0.391 (0.035)	0.381	0.403	3

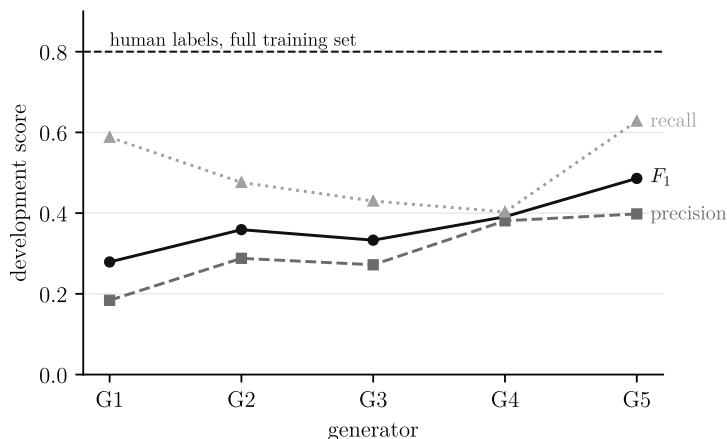


Figure 1: Development scores of each generator. The dashed line is the same classifier trained on the full training set with human labels.

Generator	F_1	P	R	Seeds
G5	0.486 (0.020)	0.398	0.628	5

3.5 Extending the generator

Each assertion is a small function that takes the sentence’s drugs and returns what the sentence must say, which pairs are positive and with which label, and which drugs it concerns. A variant applied to 26% of positive assertions retargets the assertion at something outside the sentence’s drug list, such as an endogenous substance, an unnamed drug, food or a laboratory measurement. The sentence still reads like an interaction statement, but none of its drug pairs interact, so it is a hard negative. A new assertion inherits this variant automatically, so its wording never guarantees a positive label. Adding a dedicated assertion for protein-binding displacement left G5’s uncertainty coefficient at 0.458, which can be checked before any text is generated.

Several G5 assertions encode the annotation guidelines’ rules for awkward cases. The guidelines annotate an affirmed interaction even when the same sentence also denies it (§4.5.4 of the guidelines). Their example is “Although this interaction has not been reported with cinoxacin, caution should be exercised when cinoxacin is given concomitantly with caffeine-containing products”, annotated ADVISE between caffeine and the second mention of cinoxacin. G1 to G3 had no construction for cases like this. They were added for G4, when the frames were rebuilt from the guidelines’ rules and examples.

Example 3.4

Tutorial §2.5 shows what an assertion returns, adds the displacement assertion and measures it without generating any text. On its own the assertion gives a positive sentence about three times in four, and every NONE case comes from the retargeting variant. The same section shows how the effect, denial and contradictory assertions behave alone and next to a second assertion.

→ TUTORIAL §2.5

3.6 Other observations on generators

All scores in this section are on the development set. G3 moved closer to the corpus on every distributional measure we computed, and its F_1 fell from G2’s 0.359 to 0.333. Generating G5’s sentences with the model’s high reasoning setting scored 0.453, against 0.462 with its low setting, for training sets of equal size (0.5 standard errors), at several times the cost per request. A union

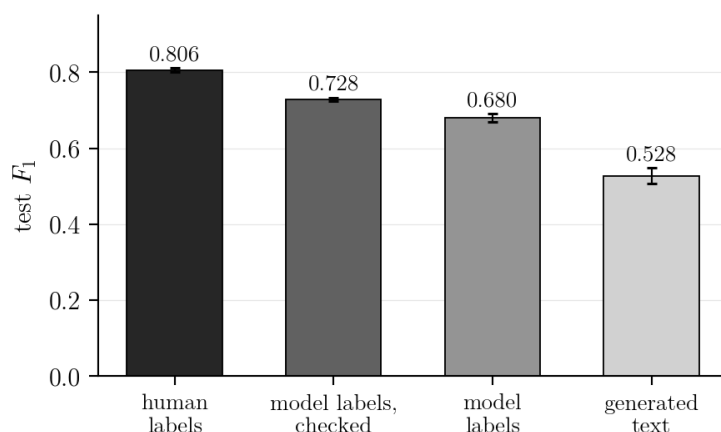


Figure 2: Test F_1 by the source of the training labels and text. The first three bars use the same 2,402 real sentences.

of G2 and G5 sentences scored 0.530, against 0.484 for G5 alone at the same number of sentences (4.6 standard errors), and doubling the size of the union scored 0.529.

4 Labelling Real Sentences

The labelling approach keeps the real training sentences and their marked drug mentions, discards the gold labels, and asks the same model to label every pair. The prompt gives the five labels and a simplified version of the guideline conventions, among them that a pair is NONE unless both drugs take part in the stated interaction, and that a denied or unstudied interaction is NONE. Each request carries eight pairs, at temperature 0 and with the model’s high reasoning setting. Requests that carried whole sentences lost 26% of pairs, most often from the sentences with the most drugs, which is why pairs are sent in groups of eight. Of 2,417 sentences, 2,402 were labelled completely.

All scores in the next paragraph are on the test set. Training on the 2,402 sentences with human labels and on the same sentences with the model’s labels gives two classifiers whose training data share text, drug mentions and pairs and differ only in their labels. Replacing the human labels with the model’s labels cost 0.126 F_1 (0.806 against 0.680, 23.7 standard errors). Replacing the real sentences with generated ones cost a further 0.152 (0.680 against 0.528, 14.7 standard errors), although the generated set has 2.3 times as many training pairs (Figure 2). An earlier comparison on the development set, with 2,248 sentences in each arm drawn independently, gave the same ordering, with 0.781 for human labels, 0.661 for model labels and 0.444 for generated text.

Against gold, the model’s labels on the training set have precision 0.598 and recall 0.873. The model finds most annotated interactions, and 36% of its positive labels are NONE in the gold annotation. Some of these errors break a guideline convention rather than misread the pharmacology, as when a drug is named twice and the model labels both mentions, while the guidelines annotate only the one inside the interaction statement.

Example 4.1

Tutorial §3 shows a real sentence that names lithium twice. The model labels both acetazolamide–lithium pairs MECHANISM, while the gold annotation marks only the mention inside “increases lithium excretion”.

→ TUTORIAL §3

5 Checking the Model’s Labels

Most of the model’s labelling errors are false positives, so only its positive labels are checked. A second call to the same model, the `verifier`, receives the sentence with every drug mention numbered and, for each positive pair, the question of whether the guidelines would annotate an interaction between those two mentions. Mentions are numbered rather than named because the same drug named twice is two mentions. A pair the verifier rejects is changed to NONE. On human-labelled development data, the verifier recalled 0.920 of NONE pairs and 0.948 of positive pairs. Numbering the mentions and adding the guideline rules to its prompt together raised its NONE recall from 0.532 to 0.902, excluding pairs of the same drug.

The verifier rejected 614 of the model’s 3,573 positive labels on the training set, and 174 requests returned errors, leaving the 474 positive labels in those sentences unchecked. The rejections target errors. Of the rejected labels, 561 (91%) are NONE in gold, against 493 of the 2,485 accepted labels (20%). Checking therefore corrected 561 of the 1,286 positive labels that gold marks NONE, and removed 53 annotated interactions. On the labels themselves, it raised precision from 0.598 to 0.707 and lowered recall from 0.873 to 0.855. The 474 unchecked labels are NONE in gold 49% of the time, a higher error rate than among the labels that were checked.

Example 5.1

Tutorial §4 prints the verifier’s exact question for the lithium sentence, its verdicts, and the gold label of every rejected and accepted positive on a 400-sentence sample.

→ [TUTORIAL §4](#)

All scores in this paragraph are on the test set. The classifier trained on checked labels has precision 0.676 and recall 0.790, against 0.578 and 0.824 for the classifier trained on unchecked labels, and its F_1 is 0.728 against 0.680 (9.5 standard errors). Checking recovered 39% of the cost of replacing human labels with model labels. On the validation set, the gain depended on which positives were removed rather than how many. Removing positives at random until the model-labelled data matched the corpus’s positive rate gained 0.013 (1.1 standard errors), and demoting the positives the verifier rejected gained 0.094. The gain from checking was 0.079 on the validation set and 0.049 on the test set, a reduction to be expected once a configuration has been chosen on validation.

All scores in this paragraph are on the development and validation sets. We also applied the verifier to generated data, where the pairs it rejects are removed rather than relabelled. Against removing the same number of pairs at random, this gained 0.041 for G2 and 0.042 for G3 on the development set, and 0.011 for G5 on the validation set (1.1 standard errors). All 878 pairs it removed from G5 had been specified as NONE, and they came disproportionately from the retargeting variant. It removed 4.8% of the pairs in sentences with that variant and 1.9% of pairs in sentences without it. On G5 the verifier mostly removed intended hard negatives, so checking gave no clear gain on that generator.

6 Spending Annotation

Table 4 gives the test scores of the configurations chosen on validation at three levels of human annotation, together with the configurations they are compared against (Figure 3). With no human annotation, checked model labels reached 0.728, 90% of the score with every training sentence human-labelled, and came within 0.02 of 1,000 human-labelled sentences alone (0.017 lower, 1.4 standard errors). With 1,000 human-labelled sentences, 42% of the training set, and checked model labels on the remaining sentences, the classifier reached 0.773, 96%. That is 0.028 above the 1,000 sentences alone (2.2 standard errors) and 0.032 below full annotation (6.2 standard errors). The 1,000-sentence human arm varied two to five times more across seeds than the arms using checked labels, because each seed drew a different set of sentences.

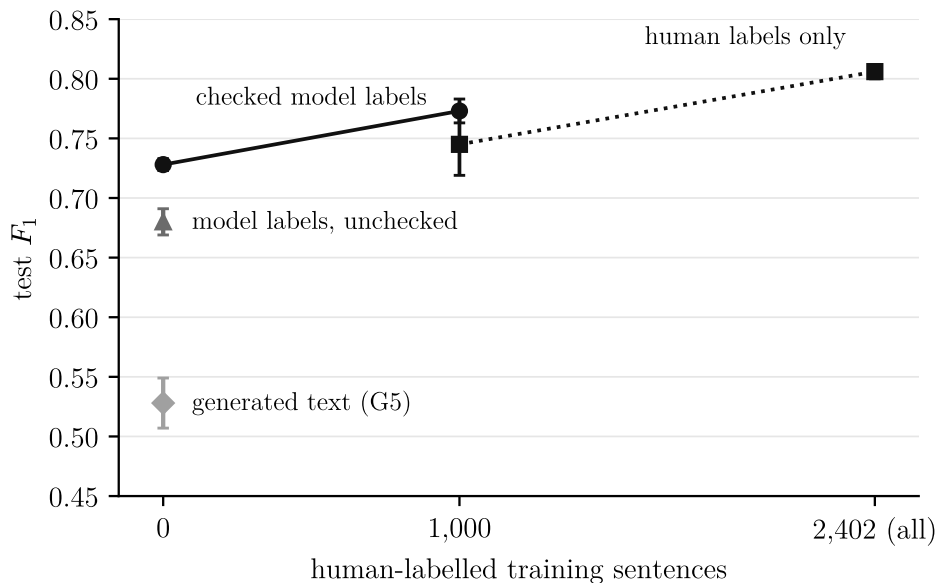


Figure 3: Test F_1 against the number of human-labelled training sentences. Bars show one standard deviation across five seeds.

Table 4: Test scores, five seeds each. Share is relative to all 2,402 training sentences with human labels.

Training data	Human sentences	F_1	s.d.	P	R	Share
Generated text (G5)	0	0.528	0.021	0.465	0.611	65.5%
Model labels	0	0.680	0.011	0.578	0.824	84.4%
Model labels, checked	0	0.728	0.005	0.676	0.790	90.4%
Human labels	1,000	0.745	0.026	0.773	0.720	92.5%
Human labels and checked model labels	1,000	0.773	0.010	0.742	0.807	96.0%
Human labels	2,402	0.806	0.006	0.797	0.815	100%

The configurations in Table 4 were chosen from a grid on the validation set, in which human budgets of 0 to 1,000 sentences were combined with each possible addition (Figure 4). The additions were nothing, the model’s labels on the remaining sentences, checked model labels, G5’s generated text, and model labels together with G5’s text. Checked model labels were the best addition at every budget. Generated text added 0.072 to 100 human-labelled sentences (2.8 standard errors) and nothing from 250 sentences upwards. Adding it to model labels lowered F_1 at every budget, by between 0.015 and 0.026.

G5’s generated text scored 0.063 higher on the test set than on the validation set. The test set is 88% DrugBank text, the kind of text G5 imitates, and on the development set G4 scored 0.396 on DrugBank sentences and 0.144 on MedLine sentences.

Example 6.1

Tutorial §5 prints the validation grid and the test table from the project’s run records.

→ TUTORIAL §5

7 Applying This to Another Task

Most of the pipeline does not depend on drug interactions. Specifications rendered into prompts, the separation of label-carrying assertions from label-neutral modifiers, stored model output with instances built from it afterwards, the uncertainty coefficient and the bag-of-words test,

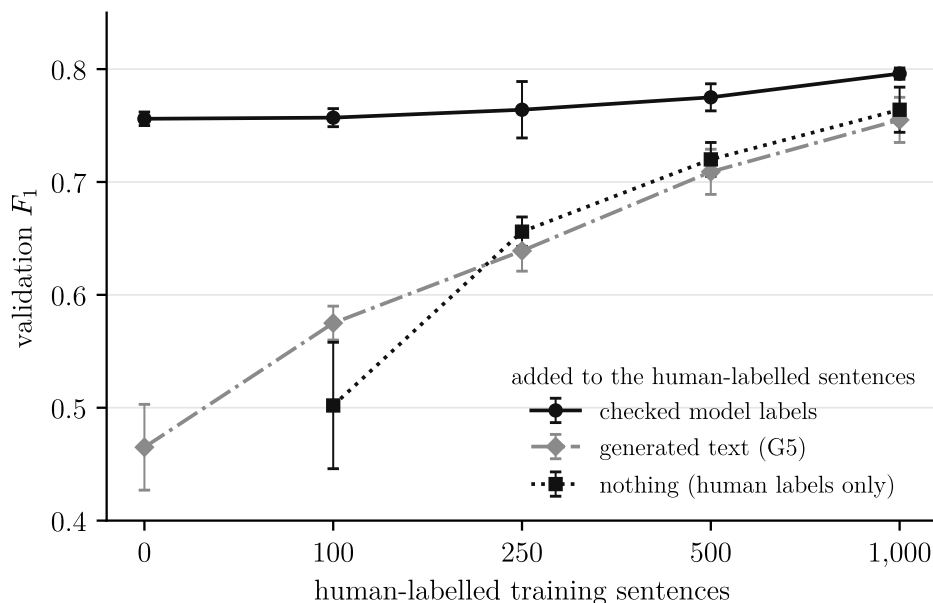


Figure 4: Validation F_1 for three additions to a human budget of 0 to 1,000 sentences. The two additions not shown, unchecked model labels and model labels with generated seeds, scored below checked model labels at every budget. Bars show one standard deviation across seeds.

the verifier’s numbered-mention question, the check of a filter against gold, and the budget grid with a single test evaluation all carry over unchanged. What is specific to DDI is the label set, the guideline conventions in the assertions and in the labelling and checking prompts, the drug vocabulary, and the pair structure.

Three practices follow from the results above.

A generator’s design should be checked before any text is generated. G4’s frames each decided their sentence’s label ($U = 1.000$), a bag-of-words model recovered the frame from masked text for 0.961 of sentences, and G4 scored 0.391 on the development set. Separating assertions from modifiers brought the coefficient to 0.458, close to the corpus’s 0.464, and F_1 rose to 0.486, almost all of it in recall. Both checks cost nothing beyond the specifications and a classified sample of real sentences, and they should be rerun after every change, since G4’s fixed descriptions of drugs outside the relation removed one cue and created another.

Comparisons across training sets of different sizes should train each to a minimum number of optimisation steps rather than a fixed number of epochs. Under a fixed three epochs, the human-only arm with 100 sentences received about 60 steps and predicted NONE for every pair in four of five seeds, and against that baseline generated text appeared to add 0.52 F_1 on the development set. With a minimum of 500 steps it added 0.072 on the validation set.

A filter should be judged by checking what it removes against gold labels. On real sentences, 91% of the verifier’s rejections were errors, against 20% of what it accepted, and it raised the classifier’s F_1 on the test set by 0.049. On G5 the same verifier removed deliberate hard negatives at 2.5 times the rate of other pairs and gave no clear gain. The same instrument helped in one setting and not the other, and only the comparison with gold showed why.

Example 7.1

Tutorial §6 lists which parts of the pipeline and code carry over to another relation extraction task, which are specific to DDI, and the lessons above.

→ TUTORIAL §6

8 How the Approach Developed

Each stage of the project tested a working assumption about why generated data scored lower than human-labelled data (Table 5). The comparison with labelling real text, which changed the direction of the project, was run in week 7.

Table 5: Working assumptions and the experiments that tested them. Week 1 began on 13 July 2026.

Week	Assumption	Test	Outcome
2–3	class balance limits generated data	NONE ratio varied from 0.3 to 0.85	F_1 between 0.25 and 0.28 on the development set at every ratio
3	generated data loses through quantity or balance	human data cut to the generated set’s size and NONE ratio	human data scored 0.460 on the development set, against 0.25 to 0.28 for generated data
4	the number of pairs predicts the label	positive rate by pairs per sentence	confirmed (74.8% against 0.2%), G2 built
5	generated data complements small human budgets	human budgets with generated data added, fixed epochs	an apparent gain of 0.52 at 100 sentences, from a baseline that predicted NONE
5	matching the corpus distribution raises F_1	G3	every distance to the corpus fell, and F_1 fell from 0.359 to 0.333
6	wrong labels explain the gap	verifier removal against random removal	gains of 0.041 for G2 and 0.042 for G3
6–7	missing constructions explain the gap	G4, then G5	the construction shortcut found ($U = 1.000$) and removed, F_1 from 0.391 to 0.486
7	labelling real text beats generating it	the same-sentence comparison of Section 4	confirmed, 0.661 for model labels against 0.444 for generated text
after 8	checking labels closes most of the gap	validation grid, then test	0.728 on the test set with no human labels, against 0.806 with all

9 Limitations

The study uses one corpus, one model for generation, labelling and checking, and one classifier, and every configuration relies on the gold drug mentions, so each assumes a separate named entity recogniser. Without negative instance filtering, our scores cannot be compared with published work. Configurations were run with three to five seeds, and repeated runs of the same design differed by up to 0.02, so smaller differences at three seeds should not be read as effects. The corpus value of the uncertainty coefficient rests on construction labels assigned by the model, not by annotators. The verifier was calibrated on the development set, and the separate effects of numbering mentions and of adding the guideline rules on its NONE recall were not measured. By the model’s construction labels, 21% of the corpus sentences that gold marks NONE contain constructions that usually indicate an interaction, and none has been checked as a possible annotation error. The test set is 88% DrugBank text.

10 Discussion

For drug-drug interaction extraction, an LLM was more useful as an annotator of real text than as a writer of new text. Replacing human labels with the model’s labels cost 0.126 F_1 on the test set, and replacing the real sentences with generated ones cost a further 0.152, although the generated set had more than twice as many training pairs. Ding et al. (2023) reached the

same ordering with GPT-3, finding that labelling real data suited tasks with small label spaces. DDI-2013 has five labels, and the ordering held with a more recent model several years later. That suggests it reflects the task more than the model.

Generation has succeeded in other studies, and the comparison they make explains part of the difference. Josifoski et al. (2023) and Delmas et al. (2024) compared generated data with distantly supervised or noisily labelled real data, and generation won. Our alternatives to generated text were real sentences with human labels or with the model’s own labels, not noisily labelled data. Tang et al. (2023) found generated data comparable to the original training data on two biomedical relation tasks, but those tasks do not have DDI’s five labels or its conventions about which mention takes part in an interaction. Our grid also agrees with Li et al. (2025) at a much larger scale. They found that generated relation extraction samples could not replace human-labelled ones even at eight examples per relation, and that they helped less as human data grew. Here, generated text added 0.072 to 100 human-labelled sentences and nothing from 250 sentences upwards.

The main obstacle to generation was not wrong labels but shortcuts in the text. In the first generator the number of drugs in a sentence predicted its label, and in the fourth the sentence’s construction determined it, while a bag-of-words model recovered that construction from masked text for 0.961 of sentences. This is the generated-data counterpart of the annotation artefacts that Gururangan et al. (2018) found in natural language inference data, where a text classifier given only the hypothesis predicted the label for about two-thirds of SNLI examples, because crowd workers writing to a requested label left consistent traces such as negation. An LLM writing to a requested label is a similar task. Classifiers trained on such data adopt the shortcuts rather than the task (McCoy et al., 2019; Geirhos et al., 2020). The uncertainty coefficient and the masked bag-of-words test give a way to find these shortcuts before generating text.

Li et al. (2025) treated the same problem as one of diversity. They generated relation extraction samples one at a time with a few labelled demonstrations in the prompt, instructed the model to make each new sample differ from the demonstrations, and fine-tuned the generating model with direct preference optimisation to favour samples that were both diverse and correct. These steps lowered the similarity between generated samples and raised downstream scores. Without them the model imitated its demonstrations, at times repeating earlier samples word for word, and our generator showed a related failure when it copied the wording of its instruction into a sentence (Appendix B). Our generators used no demonstrations and were not fine-tuned, so their diversity came from the specifications instead, through randomly drawn drug names, independently drawn modifiers and sampled instructions on how to write each sentence. Their measures of diversity, the similarity and word overlap between samples, do not involve the label. A set of sentences can vary in wording while a construction still decides each label, and that is what the uncertainty coefficient and the masked bag-of-words test measure. The two kinds of measure are complementary. Preference optimisation was not tested here. Because the weights of `gpt-oss-120b` are public, fine-tuning it to prefer sentences whose form does not reveal their label is a possible next step.

Checking the model’s positive labels worked on real text and not on generated text. On real sentences, 91% of the labels the verifier rejected were NONE in gold, against 20% of those it accepted, and training on the checked labels raised F_1 on the test set from 0.680 to 0.728. This agrees with Gero et al. (2023), who found that asking an LLM to verify its own extractions improved clinical information extraction. On G5, the same verifier mostly removed sentences written deliberately to look like interactions without being one, and gave no clear gain. A check of this kind helps only where the data contains the errors it was designed to find, and only a comparison with gold labels shows which case applies. Combined with human labels, checked model labels also behaved as Wang et al. (2021) found for GPT-3 labels, adding 0.028 to 1,000 human-labelled sentences.

The annotation guidelines mattered at every stage. Some of the model’s labelling errors

broke a guideline convention, such as which of two mentions of the same drug takes part in an interaction, rather than misreading the pharmacology. Numbering mentions and adding the guideline rules to the verifier’s prompt raised its recall of NONE pairs from 0.532 to 0.902, in line with Sainz et al. (2024), who found detailed guidelines essential for LLM information extraction. The generators improved once their constructions were rebuilt from the guidelines’ own rules and examples. The guidelines also point to possible annotation errors. By the model’s construction labels, 21% of the corpus sentences that gold marks NONE contain constructions that usually indicate an interaction, and none has been checked.

Several directions remain open. Generation was never grounded in known interactions, for example by drawing pairs from DrugBank’s interaction records, which is the relation-level form of the dictionary-assisted generation that Ding et al. (2023) tested. The cost of each approach was not measured directly, but reasoning effort points the same way as Ding et al. (2023), who found generation the more cost-effective. Generating G5’s sentences at the model’s high reasoning setting took 4.18 seconds per request and scored no better than at its low setting (0.453 against 0.462), and requests at the low setting were much faster, although they were not timed. Labelling needed the high setting. An early five-way labeller run at the low setting on human-labelled development data recalled only 0.45 to 0.48 of NONE pairs, against 0.81 at the high setting, and changes to its prompt did not close the gap. Labelling real text is therefore likely to cost more per request than generating it, even though it trains the better classifier. The comparison should be repeated on other corpora, other models and other guidelines. On the evidence of this study, a project with no annotation budget should label its real text with an LLM, check the positive labels, and spend whatever human annotation it can afford on adding to them.

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A Correspondence with the Companion Notebook

Table A1: Sections of this report and of `tutorial/ddi_tutorial.ipynb` that cover the same material.

Report	Notebook	What the notebook adds
2 Task, Data and Evaluation	§1 The task	a real sentence with repeated mentions, and the record format
3 Writing Labelled Sentences	§2 Writing labelled sentences	a G5 specification, its prompt and its labelled pairs
3.2 The pair-count shortcut	§2.1–2.2	what each generator produced, and the pair-count rule
3.3 The construction shortcut	§2.3	the frame table, the coefficient by hand, the bag-of-words test and the corpus comparison
3.4 Separating assertions from modifiers	§2.4	G5 specifications and the coefficient for G4, G5 and the corpus
3.5 Extending the generator	§2.5	a new assertion measured before generation, and three existing assertions
4 Labelling Real Sentences	§3 Labelling real sentences	a sentence labelled by the annotators and by the model
5 Checking the Model’s Labels	§4 Checking the model’s labels	the verifier’s question, its verdicts, and its rejections against gold on a sample
6 Spending Annotation	§5 Spending annotation	the validation grid and the test table
7 Applying This to Another Task	§6 Applying this to another task	what carries over, what is specific to DDI, and the lessons

B G5’s Assertions, Modifiers and Variants

The examples below are sentences G5 wrote, copied from the 400-sentence sample in the companion notebook. Each is the shortest sentence in the sample that shows its row clearly, and drug names are as the model wrote them, drawn at random from the vocabulary.

The examples also show some areas where G5 falls short. The `int` sentence repeats its instruction word for word (“all that is known is that they interact”), which the system prompt forbids, `regimen_background`, `appositive` and `coordinate` need a drug that no assertion uses, so in a specification with only two drugs they are listed but add nothing. Drugs outside every assertion are given one of seven role hints with probability 0.55, then the prompt asks the model to put each in its own words, because G4 had rendered its role descriptions verbatim. This was not fully effective as the model copied the hints anyway. On a 300-specification check run before the full generation, all eight of the most repeated 4-grams in G5’s sentences were fragments of the hints, and role language appeared next to NONE pairs far more often than in the corpus, with an adjacency gap of 0.365 against 0.001.

Table B1: Positive assertions. Weight is the probability of drawing each assertion. Each binds two drugs and makes their pair positive with the label shown, unless a variant changes it (Table B3).

Assertion	Label	Weight	What the sentence must say	Example
<code>effect</code>	EFFECT	0.17	taking the two together produces a clinical consequence, with its extent	When Camphor is taken together with Antipsychotics, a new central nervous system depression occurs.

Assertion	Label	Weight	What the sentence must say	Example
effect_pd	EFFECT	0.14	a pharmacodynamic relation between the two, such as synergy or antagonism	A synergistic effect may occur between Tyrosine hydroxylase inhibitors (10 mg) and Ketanserin (5 mg).
mechanism	MECHANISM	0.14	one drug changes a pharmacokinetic property of the other, with the direction and how it was measured	Trafermin acts on Prinaberel, elevating efflux transport as measured by steady-state levels.
advise_reason	ADVISE	0.08	a recommendation about taking the two together, with its reason	Avacincaptad pegol should be taken several hours before Ajoene to avoid the possibility of hypoglycaemia, with Ajoene administered thereafter.
advise	ADVISE	0.06	a recommendation about taking the two together, with no reason	Firmonertinib administered orally is contraindicated with Thiosalicylic acid.
effect_protect	EFFECT	0.025	one drug protects against harm caused by the other	In the treatment of acute leukemia, Tavapadon protects against Idarubicin-induced thrombocytopenia, showing a protective effect, and Idarubicin is subsequently given.
int	INT	0.015	the two interact, with no detail	The data suggest that diboterminal alfa interacts with Adenylosuccinic acid, although all that is known is that they interact.
effect_failure	EFFECT	0.01	one drug makes the other lose its therapeutic effect	Surzebiclimab: during a 12-week course, Uttroside B acted on Surzebiclimab and the patients experienced loss of therapeutic effect.
contradictory	ADVISE	0.005	an interaction not reported for one drug, with advice that still applies with another (guidelines §4.5.4)	beta-Hydroxybutyric acid: it was not reported for Protectives Against Uv-Radiation, but it still applied with beta-Hydroxybutyric acid in patients, and a lower starting dose was established.

Table B2: Negative assertions. Each makes every pair it concerns NONE. Three of them concern a single drug.

Assertion	Drugs	Weight	What the sentence must say	Example
denial	2	0.13	taking the two together leaves a property unchanged	Although selegiline is coadministered with Acasunlimab, absorption shows no significant change.
study_only	2	0.08	the interaction was studied, with no result reported	The interaction of Methylumbelliferyl chitotriose and paromomycin was evaluated in a small cohort.

Assertion	Drugs	Weight	What the sentence must say	Example
drug_property	1	0.05	a property of one drug, such as inhibiting an enzyme	Betula pendula tar oil is a potent inhibitor of CYP2E1, and macrolides were started only after the others finished.
no_evidence	2	0.025	no data exist for the combination	After initiating treatment with Estradiol benzoate (0.5 mg) and thiosulfate (10 mg), no data exist for this combination, and Pyrazolone derivatives are withheld for the ensuing 4-week period.
single_drug_effect	1	0.02	an effect of one drug on its own	Imidazole derivatives produce an added hypotension, and Piclamilast serves as a comparator.
comparative	2	0.02	the two compared with each other, such as in efficacy	Although Bamipine appears superior in efficacy compared with Uterotonics, the Uterotonics are administered according to standard protocols.
lab_interference	1	0.017	one drug interferes with a laboratory test	Fluphenazine, which was started before the others and is now stopped, accompanies medicated shampoos that interfere with a urinary assay.
incompatible	2	0.005	the two are physically incompatible when mixed	Bictegravir: Dermapoietin and ferric maltol were mixed before administration, precipitation was observed in the infusion bag, Fedrilate was retained as a fallback if the first choice did not suit, and Bictegravir was started only after the others had finished.

Table B3: Modifiers and variants. Modifiers are applied independently with the probability shown and never change a label. Variants change what an assertion’s labels apply to.

Name	Probability	What it adds or changes	Example
repeated_mention	0.25	one drug is named a second time	When coadministered, Salvianolic acid A is contraindicated with Thiouracils, and Salvianolic acid A appears in the label.
regimen_background	0.18	up to two further drugs are mentioned as also present	It is established that the interaction of SCIB1 and Diphenylmethane derivatives was characterised in a single-dose study, with Minaprine also present.
dose_detail	0.15 (DrugBank), 0.35 (MedLine)	doses or concentrations in parentheses	When eprosartan (50 mg) and prolintane (100 mg) are taken together, they produce a greater elevation of transaminases.
title	0.12	a heading naming one drug, a colon, then the statement	Teicoplanin: when co-administered with Dimazole, exhibits synergistic activity.

Name	Probability	What it adds or changes	Example
appositive	0.07	a drug in the assertion is expanded with another, as in “X, such as Y”	It is established that Shea butter, such as Dimethylthiambutene, interferes with a bioassay for antibacterial levels, and that chlorpropamide shows superior binding compared with tripeleennamine.
sequential	0.065	a timing, such as a gap between the two drugs	When Guanfacine is administered orally, acting on the central nervous system, and Cefprozil is given orally after a gap, the latter’s effectiveness is reduced.
coordinate	0.04	a further drug coordinated with one in the assertion, as in “X and Y”	Appropriate doses of Beta blocking agents with Fenyramidol have not been established, and motretinide and Zolmitriptan exhibit synergistic activity.
fragment	0.03	a table row or list item instead of a full sentence	Talniflumate (oral) - Tetra(imidazole)diacuacopper (I) - Facinicline (topical)
list_target (variant)	when a list is possible	the assertion applies to each drug in a list, giving one positive pair per drug	Efinaconazole: it exhibits a synergistic effect with Epothilone D and with naftidrofuryl.
unlisted (variant)	0.26 of positive assertions	the assertion is retargeted at something outside the drug list, so its pairs are NONE	In comparison with Yellow fever vaccines, Calcium saccharate adds depressive effects to drugs of an unspecified class.